

BALLATER AGCM, AB, Canada

WMO: 715415

Lat: 55.5769N

Lon: 117.2356W

Elev: 560

StdP: 94.78

Time Zone: -7.00 (NAM)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.8	-30.6	-37.3	0.1	-33.6	-33.8	0.2	-30.6	11.1	1.3	9.9	-3.7	2.0	150	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	28.1	17.1	26.3	16.3	24.7	15.7	19.0	25.3	17.9	24.1	16.9	22.8	3.8	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	12.7	21.9	15.5	11.8	20.5	14.5	11.0	19.2	56.4	25.4	52.7	24.4	49.5	22.9	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.8	8.5	7.5	DB	-38.6	31.1	2.0	1.3	-40.0	32.1	-41.1	32.8	-42.2	33.5	-43.7	34.5	(4)
(5)			WB	-38.7	21.1	2.0	1.4	-40.1	22.1	-41.2	22.9	-42.3	23.7	-43.8	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.4	-11.1	-12.1	-5.8	2.9	11.4	14.9	16.6	15.6	10.7	3.6	-6.9	-12.3	(6)	
	DBStd	12.70	9.36	9.50	8.49	5.44	4.24	3.20	2.52	3.03	4.52	4.76	7.75	8.15	(7)	
	HDD10.0	3463	654	619	491	217	34	3	0	1	44	201	506	691	(8)	
	HDD18.3	5851	913	852	749	463	215	109	66	92	230	456	756	949	(9)	
	CDD10.0	684	0	0	0	4	79	151	204	177	66	4	0	0	(10)	
	CDD18.3	31	0	0	0	0	1	7	12	9	2	0	0	0	(11)	
	CDH23.3	676	0	0	0	4	79	136	217	176	64	0	0	0	(12)	
	CDH26.7	124	0	0	0	1	16	22	42	34	11	0	0	0	(13)	
(14) Wind	WSAvg	3.4	3.5	3.2	3.3	4.0	4.1	3.8	2.9	2.9	3.2	3.5	3.4	3.3	(14)	
(15) Precipitation	PrecAvg	476	32	22	19	19	44	75	70	61	39	27	31	30	(15)	
	PrecMax	691	88	45	38	36	109	149	113	130	118	64	66	65	(16)	
	PrecMin	287	9	7	3	3	12	33	11	18	3	8	7	6	(17)	
	PrecStd	96	22	9	9	10	28	32	32	29	26	13	18	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.4	7.2	11.8	21.5	28.7	29.3	30.2	29.8	28.0	19.7	11.7	6.3	(19)	
		MCWB	3.9	3.8	7.0	11.6	13.7	16.8	19.2	18.6	16.3	10.6	6.6	2.6	(20)	
	2%	DB	5.9	5.1	8.8	17.2	25.5	26.6	27.6	27.4	25.1	15.7	7.8	4.2	(21)	
		MCWB	3.0	2.3	4.8	7.9	12.5	15.7	18.2	18.0	14.8	9.1	4.4	1.4	(22)	
	5%	DB	4.5	3.4	6.3	14.1	23.0	24.7	25.7	25.3	22.2	13.4	5.6	2.4	(23)	
		MCWB	2.0	1.3	3.2	6.8	11.8	15.0	17.4	16.9	13.7	7.8	2.8	0.1	(24)	
	10%	DB	2.8	1.2	4.4	11.5	21.0	22.8	24.0	23.4	19.2	11.1	3.3	0.1	(25)	
		MCWB	0.9	-0.2	1.8	5.5	10.7	14.3	16.4	16.0	12.2	6.6	1.4	-1.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.1	7.9	11.4	14.8	18.6	21.3	20.1	17.6	12.0	7.0	3.0	(27)	
		MCDB	6.9	6.8	11.3	20.7	23.6	25.8	27.9	26.5	27.0	18.3	10.7	5.7	(28)	
	2%	WB	3.1	2.6	5.1	8.6	13.4	17.1	19.5	18.7	15.2	9.7	4.6	1.6	(29)	
		MCDB	5.7	4.8	8.4	15.9	22.6	24.2	25.4	25.5	23.1	14.5	7.5	3.9	(30)	
	5%	WB	2.2	1.4	3.3	7.0	12.4	16.0	18.3	17.7	14.1	8.4	2.9	0.2	(31)	
		MCDB	4.3	3.3	6.1	13.3	21.4	23.0	24.0	23.9	21.1	12.3	5.4	2.3	(32)	
	10%	WB	1.0	-0.2	1.9	5.7	11.4	14.9	17.2	16.7	12.8	7.0	1.4	-1.7	(33)	
		MCDB	2.6	1.4	4.2	11.0	19.7	20.9	22.7	22.4	18.5	10.6	3.3	0.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	11.3	11.1	11.5	14.4	11.8	12.4	13.4	13.2	10.4	9.5	9.7	(35)	
		MCDBR	10.4	11.4	11.1	16.4	18.7	16.2	16.5	16.3	18.3	14.5	11.1	12.0	(36)	
	5% WB	MCWBR	8.1	8.9	7.7	9.0	8.6	7.3	8.6	8.2	9.0	8.7	8.0	9.9	(37)	
		MCWBR	10.1	11.0	10.9	15.1	17.1	14.1	15.1	15.3	17.1	13.3	10.7	11.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.240	0.249	0.289	0.336	0.347	0.353	0.354	0.353	0.312	0.286	0.248	0.236	(40)	
		taud	2.308	2.365	2.349	2.354	2.390	2.396	2.418	2.400	2.523	2.558	2.431	2.220	(41)	
	Edn at Noon	Edn at Noon	715	855	889	887	895	891	882	859	858	806	712	611	(42)	
		Edh at Noon	50	70	91	106	109	110	106	101	78	59	45	43	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.76	1.77	3.33	4.72	5.60	6.05	5.92	4.83	3.27	1.77	0.80	0.53	(44)	
	RadStd	RadStd	0.07	0.14	0.27	0.30	0.41	0.46	0.39	0.39	0.27	0.21	0.06	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling		Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
(47)	Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page